3.9.2 Description of effects of earthquake in Richter scale

MODIFIED MERCALLI SCALE	RICHTER SCALE
I. Felt by almost no one. II. Felt by very few people.	2.5. Generally not felt, but recorded on seismometers.
III. Tremor noticed by many, but they often do not realize it is an earthquake. IV. Felt indoors by many, Feels like a truck has struck the building. V. Felt by nearly everyone: many people awakened. Swaying trees and poles may be observed.	3.5. Felt by many people
VI. Felt by all; many people run outdoors. Furniture moved, slight damage occurs. VII. Everyone runs outdoors. Poorly built Structures considerable damaged; Slight damage elsewhere.	4.5. Some local damage may occur.
VIII. Specially designed structures damaged Slightly, others collapse. IX. All buildings considerably damaged, many shift off foundations, Noticeable cracks in ground.	6.0. A destructive earthquake.
X. Many structures destroyed. Ground is badly cracked.	7.0. A major earthquake.
XI. Almost all structures fall. Very wide cracks in ground. XII. Total destruction. Waves seen on ground surfaces, objects are tumbled and tossed.	8.0. and up Great Earthquakes.

Table 3.1 The Mercalli and Richter scales

3.9.3 Causes of Earthquakes

There are many factors controlling the occurrence of the earthquake. Some of the major factors include:

1. Plate Tectonic Movements
2. Volcanic Eruptions.
3. Construction of large dams results in earthquake. Example. Koyna dam, Maharashtra.
4. Other Reasons: The nuclear explosions also release massive energy to cause tremors in the earth crust. When underground cave collapses, earthquake may occur.

3.9.4 Effects of the Earthquakes

1. Damage to buildings, roads, rails, factories, dams, bridges etc.
2. Landslides caused by earthquakes damage infrastructure.
3. Fires in the forest and urban areas.
4. Flash floods.
5. Tsunami - The high amplitude oceanic waves caused by submarine earthquake (measuring more than 7 on Richter scale). The seismic waves travel through seawater generates high sea waves. They cause severe loss of life and property. For instance, on 26th December 2004, a tsunami originating from a magnitude 8.9 earthquake in northern Sumatra killed over 1,50,000 people in countries surrounding the Indian Ocean.

3.9.5 *Distribution of earthquakes*

1. Circum-Pacific region: This region includes all the coastal areas around the Pacific Ocean. It extends through the coasts of Alaska, Aleutian Islands,

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